

Last login: Mon Jun 15 19:32:03 on ttys002
adityadave@Adityas-MacBook-Air-3 ~ % psql -d tpch
psql (18.4)
Type "help" for help.

tpch=# SET ROLE alice;
SET

```
tpch=> EXPLAIN (
  ANALYZE true,
  VERBOSE true,
  COSTS true,
  BUFFERS false,
  TIMING false,
  FORMAT JSON
)
select
  o_orderpriority,
  count(*) as order_count
from
  orders
where
  o_orderdate >= date '1994-01-01'
  and o_orderdate < date '1994-01-01' + interval '3' month
  and exists (
    select
      *
    from
      lineitem
    where
      l_orderkey = o_orderkey
      and l_commitdate < l_receiptdate
  )
group by
  o_orderpriority
order by
  o_orderpriority;
```

WARNING: ===== planner input query tree =====

```
DETAIL:  {QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs true
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity true
:hasGroupRTE true
:isReturn false
:cteList <>
:rtable (
  {RANGETBENTRY
:alias <>
:eref
  {ALIAS
:aliasname orders
:colnames ("o_orderkey" "o_custkey" "o_orderstatus" "o_totalprice" "o_orderdate" "o_orderpriority" "o_clerk" "o_shippriority" "o_comment")
}
}
:rtekind 0
:relid 16561
:inh true
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals (
  {SUBLINK
:sublinkType 0
:sublinkId 0
:testexpr <>
:operName <>
:subselect
  {QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
  {RANGETBENTRY
:alias
  {ALIAS
:aliasname c
:colnames <>
}
}
:eref
  {ALIAS
:aliasname c
:colnames ("c_custkey" "c_name" "c_address" "c_nationkey" "c_phone" "c_acctbal" "c_mktsegment" "c_comment")
}
}
:rtekind 0
:relid 16546
:inh true
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rteperminfos (
  {RTEPERMISSIONINFO
:relid 16546
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 13)
:insertedCols (b)
:updatedCols (b)
}
)
)
:jointree
  {FROMEXPR
:fromlist (
  {RANGETBLREF
:rtindex 1
}
)
:quals
  {BOOLEXP
:boolop and
:args (
  {OPEXP
:opno 96
:opfuncid 65
```

```

:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
  {VAR
    :varno 1
    :varattno 1
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 1
    :location -1
  }
  {VAR
    :varno 1
    :varattno 2
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 1
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 2
    :location -1
  }
)
:location -1
}
{OPEXPR
:opno 674
:opfuncid 297
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
  {VAR
    :varno 1
    :varattno 6
    :vartype 701
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 6
    :location -1
  }
  {FUNCEXP
    :funcid 316
    :funcresulttype 701
    :funcset false
    :funcvariadic false
    :funcformat 2
    :funccollid 0
    :inputcollid 0
    :args (
      {CONST
        :consttype 23
        :consttypmod -1
        :constcollid 0
        :constlen 4
        :constbyval true
        :constisnull false
        :location -1
        :constvalue 4 [ -120 19 0 0 0 0 0 0 ]
      }
    )
    :location -1
  }
)
:location -1
}
:location -1
}
:location -1
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
  {TARGETENTRY
    :expr (
      {CONST
        :consttype 23
        :consttypmod -1
        :constcollid 0
        :constlen 4
        :constbyval true
        :constisnull false
        :location -1
        :constvalue 4 [ 1 0 0 0 0 0 0 0 ]
      }
    )
    :resno 1
    :resname ?column?
    :ressortgroupref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
  }
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}
:location -1
}
}
{RANGETBENTRY
:alias <>
:eref (
  {ALIAS
    :aliasname *GROUP*
    :colnames ("o_orderpriority")
  }
)
:rtekind 9
:groupexprs (
  {VAR
    :varno 1
    :varattno 6
    :vartype 25
    :vartypmod -1

```

c.c_custkey = orders.o_custkey

c.c_acctbal > 5000

```

:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location 116
}
)
:lateral false
:inFromCl false
:securityQuals <>
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 10561
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 12 13)
:insertedCols (b)
:updatedCols (b)
}
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
)
:quals
{BOOLEXP
:boolop and
:args (
{OPEXP
:opno 1098
:opfuncid 1090
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 5
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location 107
}
{CONST
:consttype 1082
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location 207
:constvalue 4 [ 113 -9 -1 -1 -1 -1 -1 ]
}
)
:location 199
}
{OPEXP
:opno 2345
:opfuncid 2338
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 5
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location 228
}
{OPEXP
:opno 1076
:opfuncid 2071
:opresulttype 1114
:opretset false
:opcollid 0
:inputcollid 0
:args (
{CONST
:consttype 1082
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location 247
:constvalue 4 [ 113 -9 -1 -1 -1 -1 -1 ]
}
{CONST
:consttype 1186
:consttypmod 196607
:constcollid 0
:constlen 16
:constbyval false
:constisnull false
:location 271
:constvalue 16 [ 0 0 0 0 0 0 0 0 0 0 0 0 3 0 0 0 ]
}
)
:location 260
}
)
:location 240
}
{SUBLINK
:subLinkType 0
:subLinkId 0
:testexpr <>
:operName <>
:subselect
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStat <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false

```

o_orderdate >= DATE
'1994-01-01'

o_orderdate < DATE '1994-01-01' +
INTERVAL '3 month'

```

:cteList <>
:rtable (
{RANGETBLENTTRY
:alias <>
:eref
:alias
:aliasname lineitem
:colnames ("l_orderkey" "l_partkey" "l_suppkey" "l_linenu
mber" "l_quantity" "l_extendedprice" "l_discount" "l_tax"
"l_returnflag" "l_linestatus" "l_shipdate" "l_commitdate
" "l_receiptdate" "l_shipinstruct" "l_shipmode" "l_commen
t")
}
:rtekind 0
:relid 16566
:inh true
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 16566
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22
23)
:insertedCols (b)
:updatedCols (b)
}
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rindex 1
}
)
:quals
{BOOLEXP
:boolop and
:args (
{OPEXP
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 1
:varitype 23
:varitypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 387
}
{VAR
:varno 1
:varattno 1
:varitype 23
:varitypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 1
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 400
}
)
:location 398
}
{OPEXP
:opno 1095
:opfuncid 1087
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 12
:varitype 1082
:varitypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 12
:location 427
}
{VAR
:varno 1
:varattno 13
:varitype 1082
:varitypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 13
:location 442
}
)
:location 440
}
)
:location 423
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 1
:varitype 23
:varitypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 325
}
}
:resno 1
:resname l_orderkey
:ressortgroupref 0

```

l.l_orderkey =
o.o_orderkey

l.l_commitdate <
l.l_receiptdate

```

:resorigtbl 16566
:resorigcol 1
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location 325
}
:resno 2
:resname 1_partkey
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 2
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location 325
}
:resno 3
:resname 1_suppley
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 3
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 4
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 4
:location 325
}
:resno 4
:resname 1_linenummer
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 4
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 5
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location 325
}
:resno 5
:resname 1_quantity
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 5
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 6
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location 325
}
:resno 6
:resname 1_extendedprice
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 6
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 7
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 7
:location 325
}
:resno 7
:resname 1_discount
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 7
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 8
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1

```

```

:varattnosyn 8
:location 325
}
:resno 8
:resname 1_tax
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 8
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 9
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 9
:location 325
}
:resno 9
:resname 1_returnflag
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 9
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 10
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 10
:location 325
}
:resno 10
:resname 1_linestatus
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 10
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 11
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 11
:location 325
}
:resno 11
:resname 1_shipdate
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 11
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 12
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 12
:location 325
}
:resno 12
:resname 1_commitdate
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 12
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 13
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 13
:location 325
}
:resno 13
:resname 1_receiptdate
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 13
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 14
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 14
:location 325
}
:resno 14
:resname 1_shipinstruct
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 14
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 15
:vartype 25

```

```

        :vartypmod -1
        :varcollid 100
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 15
        :location 325
    }
    :resno 15
    :resname l_shipmode
    :ressortgroupref 0
    :resorigtbl 16566
    :resorigcol 15
    :resjunk false
    }
    {TARGETENTRY
    :expr
    {VAR
    :varno 1
    :varattno 16
    :vartype 25
    :vartypmod -1
    :varcollid 100
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 16
    :location 325
    }

    :resno 16
    :resname l_comment
    :ressortgroupref 0
    :resorigtbl 16566
    :resorigcol 16
    :resjunk false
    }
    )
    :override 0
    :onConflict <>
    :returningOldAlias <>
    :returningNewAlias <>
    :returninglist <>
    :groupClause <>
    :groupDistinct false
    :groupingSets <>
    :havingQual <>
    :windowClause <>
    :distinctClause <>
    :sortClause <>
    :limitOffset <>
    :limitCount <>
    :limitOption 0
    :rowMarks <>
    :setOperations <>
    :constraintDecls <>
    :withCheckOptions <>
    :stmt_location 0
    :stmt_len 0
    }
    :location 289
    }
    )
    :location 224
    }
    }
    :mergeActionList <>
    :mergeTargetRelation 0
    :mergeJoinCondition <>
    :targetList (
    {TARGETENTRY
    :expr
    {VAR
    :varno 2
    :varattno 1
    :vartype 25
    :vartypmod -1
    :varcollid 100
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 2
    :varattnosyn 1
    :location -1
    }
    :resno 1
    :resname o_orderpriority
    :ressortgroupref 1
    :resorigtbl 16561
    :resorigcol 0
    :resjunk false
    }
    {TARGETENTRY
    :expr
    {AGGREF
    :aggfnoid 2803
    :aggtype 20
    :aggcollid 0
    :inputcollid 0
    :aggtranstype 0
    :aggargtypes <>
    :aggdirectargs <>
    :args <>
    :aggorder <>
    :aggdistinct <>
    :aggfilter <>
    :aggstar true
    :aggvarladic false
    :aggkind n
    :aggpresorted false
    :agglevelsup 0
    :aggssplit 0
    :aggno -1
    :aggtransno -1
    :location 137
    }
    :resno 2
    :resname order_count
    :ressortgroupref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
    }
    )
    :override 0
    :onConflict <>
    :returningOldAlias <>
    :returningNewAlias <>
    :returninglist <>
    :groupClause (
    {SORTGROUPCLAUSE
    :tleSortGroupRef 1
    :eqop 98
    :sorttop 664
    :reverse_sort false
    :nulls_first false
    :hashable true
    }
    )
    :groupDistinct false
    :groupingSets <>
    :havingQual <>
    :windowClause <>
    :distinctClause <>
    :sortClause (
    {SORTGROUPCLAUSE
    :tleSortGroupRef 1

```

```

:eqop 98
:sortop 664
:reverse_sort false
:nulls_first false
:hashable true
}
)
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 0
}

WARNING: pull_up_sublinks - File: src/backend/optimizer/prop/prepjointree.c, Relation: N/A, Explanation: Initiating pull-up of sublinks (ANY/EXISTS)
WARNING: ===== after pull up and before planning path list =====
DETAIL: {QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:ctelist <>
:rtable (
{RANGETBLENTY
:alias
{ALIAS
:aliasname c
:colnames <>
}
:eref
{ALIAS
:aliasname c
:colnames ("c_custkey" "c_name" "c_address" "c_nationkey" "c_phone" "
c_acctbal" "c_mktsegment" "c_comment")
}
:rtekind 0
:relid 16546
:inh false
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 16546
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 13)
:insertedCols (b)
:updatedCols (b)
}
)
}
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtdindex 1
}
)
}
:quals (
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
{PARAM
:paramkind 1
:paramid 0
:paramtype 23
:paramtypmod -1
:paramcollid 0
:location -1
}
)
:location -1
}
)
{OPEXPR
:opno 674
:opfuncid 297
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 6
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
{CONST
:consttype 701
:consttypmod -1
:constcollid 0
:constlen 8
:constbyval true
:constisnull false
:location -1
:constvalue 8 [ 0 0 0 0 0 -120 -77 64 ]
}
)
:location -1
}
}

```

l.l_orderkey =
o.o_orderkey

c.c_acctbal > 5000

```

    )
  }
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList <>
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}

WARNING: ===== after pull up and before planning path list =====
DETAIL: {QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
  {RANGETBENTRY
    :alias
      {ALIAS
        :aliasname c
        :colnames <>
      }
    :eref
      {ALIAS
        :aliasname c
        :colnames ("c_custkey" "c_name" "c_address" "c_nationkey" "c_phone" "
          c_acctbal" "c_mktsegment" "c_comment")
      }
    :rtekind 0
    :relid 16546
    :inh false
    :relkind r
    :rellockmode 1
    :perminfoindex 1
    :tablesample <>
    :lateral false
    :inFromCl true
    :securityQuals <>
  }
)
:rteperminfos (
  {RTEPERMISSIONINFO
    :relid 16546
    :inh true
    :requiredPerms 2
    :checkAsUser 0
    :selectedCols (b 8 13)
    :insertedCols (b)
    :updatedCols (b)
  }
)
:jointree
  {FROMEXPR
    :fromlist (
      {RANGETBLREF
        :rtindex 1
      }
    )
  }
:quals (
  {OEXPR
    :opno 674
    :opfuncid 297
    :opresulttype 16
    :opretset false
    :opcollid 0
    :inputcollid 0
    :args (
      {VAR
        :varno 1
        :varattno 6
        :vartype 701
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 6
        :location -1
      }
      {CONST
        :consttype 701
        :consttypmod -1
        :constcollid 0
        :constlen 8
        :constbyval true
        :constisnull false
        :location -1
        :constvalue 8 [ 0 0 0 0 0 -120 -77 64 ]
      }
    )
    :location -1
  }
)
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
  {TARGETENTRY
    :expr
      {VAR
        :varno 1
        :varattno 1
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 1
        :location -1
      }
    :resno 1
  }
)

```

c.c_acctbal > 5000

```

:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}

```

WARNING: ===== after pull up and before planning path list =====

```

DETAIL: {QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs true
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity true
:hasGroupRTE true
:isReturn false
:cteList <>
:rtable (
{RANGETBLENTTRY
:alias <>
:eref
{ALIAS
:aliasname orders
:colnames ("o_orderkey" "o_custkey" "o_orderstatus" "o_totalprice" "o_orderdate" "o_orderpriority" "o_clerk" "o_shippriority" "o_comment")
}
:rtekind 0
:relid 16561
:inh false
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals ((
{ALTERNATIVESUBPLAN
:subplans (
{SUBPLAN
:subLinkType 0
:testexpr <>
:paramIds <>
:plan_id 1
:plan_name SubPlan\ 1
:firstColType 2278
:firstColTypmod -1
:firstColCollation 0
:useHashTable false
:unknownEqFalse true
:parallel_safe false
:setParam <>
:parParam (i 0)
:args (
{VAR
:varno 1
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location -1
}
)
:startup_cost 0
:per_call_cost 5922
}
{SUBPLAN
:subLinkType 2
:testexpr
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location -1
}
}
{PARAM
:paramkind 1
:paramid 1
:paratype 23
:paratypmod -1
:paramcollid 0
:location -1
}
)
:location -1
}
}
)
:paramIds (i 1)
:plan_id 2
:plan_name SubPlan\ 2
:firstColType 23
:firstColTypmod -1
:firstColCollation 0
:useHashTable true
:unknownEqFalse true
:parallel_safe true
:setParam <>
:parParam <>
}
}

```

c.c_custkey =
orders.o_custkey

```

:args <>
:startup_cost 5716.52
:per_call_cost 0.0025
}
}
})
}
{RANGETBLENTY
:alias <>
:eref
{ALIAS
:aliasname *GROUP*
:colnames ("o_orderpriority")
}
:rtekind 9
:groupexprs (
{VAR
:varno 1
:varattno 6
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location 116
}
)
:lateral false
:inFromCl false
:securityQuals <>
}
{RANGETBLENTY
:alias <>
:eref
{ALIAS
:aliasname lineitem
:colnames ("l_orderkey" "l_partkey" "l_suppkey" "l_linenum" "l_quantity" "l_extendedprice" "l_discount" "l_tax" "l_returnflag" "l_lines" "l_shipdate" "l_commitdate" "l_receiptdate" "l_shipinstruct" "l_shipmode" "l_comment")
}
:rtekind 0
:relid 16566
:inh false
:relkind r
:rellockmode 1
:permindex 2
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 16561
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 12 13)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 16566
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23)
:insertedCols (b)
:updatedCols (b)
}
)
}
:jointree
{FROMEXPR
:fromlist (
{JOINEXPR
:jointype 4
:isNatural false
:larg
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
)
:quals (
{OEXPR
:opno 1098
:opfncid 1090
:opresulttype 16
:opreset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 5
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location 187
}
{CONST
:consttype 1082
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location 207
:constvalue 4 [ 113 -9 -1 -1 -1 -1 -1 -1 ]
}
)
:location 199
}
{OEXPR
:opno 2345
:opfncid 2338
:opresulttype 16
:opreset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 5
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location 228
}
{CONST
:consttype 1114

```

o.o_orderdate >= DATE '1994-01-01'

o.o_orderdate < '1994-04-01 00:00:00'

```

:consttypmod -1
:constcollid 0
:constlen 8
:constbyval true
:constisnull false
:location -1
:constvalue 8 [ 0 32 105 23 -25 90 -1 -1 ]
}
}
:location 240
}
}
}
:rarg
{RANGETBLREF
:rtindex 3
}
:usingClause <>
:join_using_alias <>
:quals (
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 3
:varattno 1
:varatype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 1
:location 387
}
{VAR
:varno 1
:varattno 1
:varatype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 400
}
)
:location 398
}
{OPEXPR
:opno 1095
:opfuncid 1087
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 3
:varattno 12
:varatype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 12
:location 427
}
{VAR
:varno 3
:varattno 13
:varatype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 13
:location 442
}
)
:location 440
}
}
:alias <>
:rtindex 0
}
)
:quals <>
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 6
:varatype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
:resno 1
:resname o_orderpriority
:ressortgroupref 1
:resorigtbl 16561
:resorigcol 6
:resjunk false
}
{TARGETENTRY
:expr
{AGGREF
:aggfnoid 2803
:aggtype 20
:aggcollid 0
:inputcollid 0
:aggtranstype 0
:aggargtypes <>
:aggdirectargs <>
:args <>
:aggorder <>
:aggdistinct <>
:aggfilter <>
:aggstar true
:aggvariadic false
:aggkind n
:aggpresorted false
:agglevelsup 0
:aggsplit 0
:aggno -1

```

lineitem.l_orderkey =
orders.o_orderkey

lineitem.l_commitdate <
lineitem.l_receiptdate

```

      :aggtransno -1
      :location 137
    }
  }
  :resno 2
  :resname order_count
  :ressortgroupref 0
  :resorigtbl 0
  :resorigcol 0
  :resjunk false
}
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause (
  {SORTGROUPCLAUSE
    :tleSortGroupRef 1
    :eqop 98
    :sortop 664
    :reverse_sort false
    :nulls_first false
    :hashable true
  }
)
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause (
  {SORTGROUPCLAUSE
    :tleSortGroupRef 1
    :eqop 98
    :sortop 664
    :reverse_sort false
    :nulls_first false
    :hashable true
  }
)
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 0
}

WARNING: === PLANNER SUGGESTION ===
Plan:
GroupAggregate (cost=8883274415.71..8883274719.76 rows=5 width=17)
  Output: orders.o_orderpriority, count(*)
  Group Key: orders.o_orderpriority
  -> Sort (cost=8883274415.71..8883274517.04 rows=40534 width=9)
    Output: orders.o_orderpriority
    Sort Key: orders.o_orderpriority
    -> Hash Semi Join (cost=212549.34..8883271313.47 rows=40534 width=9)
      Output: orders.o_orderpriority
      Hash Cond: (orders.o_orderkey = lineitem.l_orderkey)
      -> Seq Scan on public.orders (cost=0.00..8883048508.00 rows=144199 width=13)
        Output: orders.o_orderpriority, orders.o_orderkey
        Filter: ((orders.o_orderdate >= '1994-01-01'::date) AND (ANY (orders.o_custkey = (hashed SubPlan 2).col1)) AND (orders.o_orderdate < '1994-04-01 00:00:00'::timestamp without time zon
e))
      SubPlan 2
        -> Seq Scan on public.customer c (cost=0.00..5547.00 rows=67808 width=4)
          Output: c.c_custkey
          Filter: (c.c_acctbal > '5000'::double precision)
        -> Hash (cost=179731.50..179731.50 rows=2000307 width=4)
          Output: lineitem.l_orderkey
          -> Seq Scan on public.lineitem (cost=0.00..179731.50 rows=2000307 width=4)
            Output: lineitem.l_orderkey
            Filter: (lineitem.l_commitdate < lineitem.l_receiptdate)

=====
WARNING: === EXECUTOR EXECUTION ===
Rows Processed: 5
Plan & Stats:
GroupAggregate (cost=8883274415.71..8883274719.76 rows=5 width=17) (actual time=2605.450..2607.146 rows=5.00 loops=1)
  Output: orders.o_orderpriority, count(*)
  Group Key: orders.o_orderpriority
  Buffers: shared hit=4239 read=130164, temp read=14748 written=14748
  -> Sort (cost=8883274415.71..8883274517.04 rows=40534 width=9) (actual time=2605.009..2605.828 rows=23317.00 loops=1)
    Output: orders.o_orderpriority
    Sort Key: orders.o_orderpriority
    Sort Method: quicksort Memory: 769kB
    Buffers: shared hit=4239 read=130164, temp read=14748 written=14748
    -> Hash Semi Join (cost=212549.34..8883271313.47 rows=40534 width=9) (actual time=1820.722..2598.067 rows=23317.00 loops=1)
      Output: orders.o_orderpriority
      Hash Cond: (orders.o_orderkey = lineitem.l_orderkey)
      Buffers: shared hit=2436 read=130164, temp read=14748 written=14748
      -> Seq Scan on public.orders (cost=0.00..8883048508.00 rows=144199 width=13) (actual time=44.804..389.424 rows=25383.00 loops=1)
        Output: orders.o_orderpriority, orders.o_orderkey
        Filter: ((orders.o_orderdate >= '1994-01-01'::date) AND (ANY (orders.o_custkey = (hashed SubPlan 2).col1)) AND (orders.o_orderdate < '1994-04-01 00:00:00'::timestamp without time zon
e))
      Rows Removed by Filter: 1474617
      Buffers: shared hit=3954 read=25726
      SubPlan 2
        -> Seq Scan on public.customer c (cost=0.00..5547.00 rows=67808 width=4) (actual time=0.031..30.274 rows=67989.00 loops=1)
          Output: c.c_custkey
          Filter: (c.c_acctbal > '5000'::double precision)
          Rows Removed by Filter: 82011
          Buffers: shared hit=3672
        -> Hash (cost=179731.50..179731.50 rows=2000307 width=4) (actual time=1770.924..1770.924 rows=3793296.00 loops=1)
          Output: lineitem.l_orderkey
          Buckets: 262144 (originally 262144) Batches: 32 (originally 16) Memory Usage: 6223kB
          Buffers: shared hit=282 read=104438, temp written=10750
          -> Seq Scan on public.lineitem (cost=0.00..179731.50 rows=2000307 width=4) (actual time=0.643..1154.424 rows=3793296.00 loops=1)
            Output: lineitem.l_orderkey
            Filter: (lineitem.l_commitdate < lineitem.l_receiptdate)
            Rows Removed by Filter: 2207919
            Buffers: shared hit=282 read=104438

=====

```

QUERY PLAN

```

-----
[
  +
  {
    +
    "Plan": {
      +
      "Node Type": "Aggregate",
      +
      "Strategy": "Sorted",
      +
      "Partial Mode": "Simple",
      +
      "Parallel Aware": false,
      +
      "Async Capable": false,
      +
      "Startup Cost": 8883274415.71,
      +
      "Total Cost": 8883274719.76,
      +
      "Plan Rows": 5,
      +
      "Plan Width": 17,
      +
      "Actual Rows": 5.00,
      +
      "Actual Loops": 1,
      +
      "Disabled": false,
    }
  }
]

```

```

+
+Output": ["orders.o_orderpriority", "count(*)"],
+
+Group Key": ["orders.o_orderpriority"],
+
+Plans": [
+
+  {
+
+    "Node Type": "Sort",
+
+    "Parent Relationship": "Outer",
+
+    "Parallel Aware": false,
+
+    "Async Capable": false,
+
+    "Startup Cost": 8883274415.71,
+
+    "Total Cost": 8883274517.04,
+
+    "Plan Rows": 40534,
+
+    "Plan Width": 9,
+
+    "Actual Rows": 23317.00,
+
+    "Actual Loops": 1,
+
+    "Disabled": false,
+
+    "Output": ["orders.o_orderpriority"],
+
+    "Sort Key": ["orders.o_orderpriority"],
+
+    "Sort Method": "quicksort",
+
+    "Sort Space Used": 769,
+
+    "Sort Space Type": "Memory",
+
+    "Plans": [
+
+      {
+
+        "Node Type": "Hash Join",
+
+        "Parent Relationship": "Outer",
+
+        "Parallel Aware": false,
+
+        "Async Capable": false,
+
+        "Join Type": "Semi",
+
+        "Startup Cost": 212549.34,
+
+        "Total Cost": 8883271313.47,
+
+        "Plan Rows": 40534,
+
+        "Plan Width": 9,
+
+        "Actual Rows": 23317.00,
+
+        "Actual Loops": 1,
+
+        "Disabled": false,
+
+        "Output": ["orders.o_orderpriority"],
+
+        "Inner Unique": false,
+
+        "Hash Cond": "(orders.o_orderkey = lineitem.l_orderkey)",
+
+        "Plans": [
+
+          {
+
+            "Node Type": "Seq Scan",
+
+            "Parent Relationship": "Outer",
+
+            "Parallel Aware": false,
+
+            "Async Capable": false,
+
+            "Relation Name": "orders",
+
+            "Schema": "public",
+
+            "Alias": "orders",
+
+            "Startup Cost": 0.00,
+
+            "Total Cost": 8883048508.00,
+
+            "Plan Rows": 144199,
+
+            "Plan Width": 13,
+
+            "Actual Rows": 25383.00,
+
+            "Actual Loops": 1,
+
+            "Disabled": false,
+
+            "Output": ["orders.o_orderpriority", "orders.o_orderkey"],
+
+            "Filter": "((orders.o_orderdate >= '1994-01-01'::date) AND (ANY (orders.o_custkey = (hashed SubPlan 2).coll)) AND (orders.o_orderdate < '1994-04-01 00:00:00'::timestamp without time zone))",
+
+            "Rows Removed by Filter": 1474617,
+
+            "Plans": [
+
+              {
+
+                "Node Type": "Seq Scan",
+
+                "Parent Relationship": "SubPlan",
+
+                "Subplan Name": "SubPlan 2",
+
+                "Parallel Aware": false,
+
+                "Async Capable": false,
+
+                "Relation Name": "customer",
+
+                "Schema": "public",
+
+                "Alias": "c",
+
+                "Startup Cost": 0.00,
+
+                "Total Cost": 5547.00,
+
+                "Plan Rows": 67808,
+
+                "Plan Width": 4,
+
+                "Actual Rows": 67989.00,
+
+                "Actual Loops": 1,
+
+                "Disabled": false,
+
+                "Output": ["c.c_custkey"],

```

```

+         "Filter": "(c.c_acctbal > '5000'::double precision)",
+         "Rows Removed by Filter": 82011
+     }
+ ]
+ },
+ {
+     "Node Type": "Hash",
+     "Parent Relationship": "Inner",
+     "Parallel Aware": false,
+     "Async Capable": false,
+     "Startup Cost": 179731.50,
+     "Total Cost": 179731.50,
+     "Plan Rows": 2000307,
+     "Plan Width": 4,
+     "Actual Rows": 3793296.00,
+     "Actual Loops": 1,
+     "Disabled": false,
+     "Output": ["lineitem.l_orderkey"],
+     "Hash Buckets": 262144,
+     "Original Hash Buckets": 262144,
+     "Hash Batches": 32,
+     "Original Hash Batches": 16,
+     "Peak Memory Usage": 6223,
+     "Plans": [
+     {
+         "Node Type": "Seq Scan",
+         "Parent Relationship": "Outer",
+         "Parallel Aware": false,
+         "Async Capable": false,
+         "Relation Name": "lineitem",
+         "Schema": "public",
+         "Alias": "lineitem",
+         "Startup Cost": 0.00,
+         "Total Cost": 179731.50,
+         "Plan Rows": 2000307,
+         "Plan Width": 4,
+         "Actual Rows": 3793296.00,
+         "Actual Loops": 1,
+         "Disabled": false,
+         "Output": ["lineitem.l_orderkey"],
+         "Filter": "(lineitem.l_commitdate < lineitem.l_receiptdate)",
+         "Rows Removed by Filter": 2207919
+     }
+ ]
+ }
+ ]
+ }
+ ],
+ "Planning Time": 5.485,
+ "Triggers": [
+ ],
+ "Execution Time": 2619.169
+ }
+ ]
+ (1 row)
tpch=>

```